

Design Requirements: [Decision Matrix](#)

User Comfort :

The cart must include padded handles, support structures, and shock-absorbing wheels to minimize strain and enhance user comfort during extended use. Research on ergonomic tools highlights the role of padded supports and shock absorption in preventing long-term injuries and enhancing comfort. Mr. Kiser validated this requirement, noting that prolonged use often leads to discomfort.

Stability:

The cart must maintain stability when fully loaded, featuring a low center of gravity and wide wheelbase to prevent tipping and ensure safe maneuverability. Research in cart stability points to the importance of a low center of gravity and wide wheelbase in preventing accidents, especially when carrying heavy loads. Stakeholders agreed on the importance of this feature.

Attaching/Detaching Ease:

The cart should feature tool-free mechanisms for attaching and detaching accessories, ensuring quick and effortless configuration changes. Studies on user-friendly design highlight the importance of minimizing the need for tools to increase efficiency and ease of use. Mr. Kiser validated this requirement, highlighting the need for quick adjustments. Feedback included ensuring the mechanisms are intuitive and require minimal force to operate.

Adjustability:

The cart must feature adjustable handles and platform height to accommodate users of various heights and ergonomic preferences, ensuring a customizable user experience. This requirement ties to research on

ergonomic design, which emphasizes the importance of adjustability in reducing the risk of musculoskeletal disorders. After presenting this requirement to Mr. Kiser, he agreed that adjustability would significantly improve comfort and usability.

Compactness:

The cart should have a foldable design, allowing it to be easily stored in small spaces when not in use without compromising structural integrity. Studies on space-saving solutions in equipment design emphasize the practicality of foldable designs in environments with limited storage. Mr. Kiser validated this requirement, highlighting the limited storage space available.

Appeal to the Eye:

The cart should have a sleek, modern design with customizable color options to appeal aesthetically to a diverse user base. Research on product design aesthetics shows that visually appealing designs enhance user satisfaction and acceptance. Students and staff appreciated the focus on aesthetics.